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ECE 486: Final Project Report

Reaction Wheel Pendulum

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Introduction

The Reaction Wheel Pendulum (RWP) is a single-link pendulum with a motor-driven reaction wheel mounted at the free end. The system is instrumented with two optical encoders: one mounted to the fixed base that measures the pendulum angle θ_p (5000 ticks/rev), and a second on the motor shaft that measures the rotor angle θ_r relative to the pendulum (4000 ticks/rev). Actuation is provided by a single 24 V permanent-magnet DC motor driving the reaction wheel; by Newton's third law, the torque on the wheel produces an equal and opposite reaction torque on the pendulum, which is the lone mechanism by which the pendulum can be controlled. The system has two equilibria: the hanging $(\theta_p = 0)$ configuration, which is stable, and the inverted $(\theta_p = \pi)$ configuration, which is unstable and is the equilibrium we stabilize in this project. Controllers were implemented in MATLAB/Simulink and deployed to the C2000 DAQBoard via the C2000 Microcontroller Blockset.

Mathematical Model

Using the Lagrangian method with generalized coordinates $q_1 = \theta_p$ and $q_2 = \theta_r$, the kinetic energy of the system is $T = \frac{1}{2}J\dot{\theta}_p^2 + \frac{1}{2}J_r(\dot{\theta}_p + \dot{\theta}_r)^2$, where $J = J_p + m_p\ell_p^2 + m_r\ell_r^2$ is the total moment of inertia of the pendulum-plus-stator structure about the pivot. The potential energy, measured from the bottom of the swing, is $V = mg\ell(1 - \cos\theta_p)$, with $m\ell = m_p\ell_p + m_r\ell_r$. Substituting these into Lagrange's equations and noting that the motor torque $\tau = ki$ acts on the rotor $(+\tau)$ and reacts on the pendulum $(-\tau)$, the equations of motion reduce to

$$J\,\ddot{\theta}_p + J\,\omega_{np}^2 \sin\theta_p = -\tfrac{k}{\ell}\,u, \qquad \ddot{\theta}_r = \tfrac{k}{J_r}\,u, \tag{1}$$

where $\omega_{np}^2 = g/\ell$ is the small-oscillation frequency of the pendulum about its hanging position and the control input u is scaled so $|u| \le 10$ (10 units = max motor current). Defining the lumped gains $a = \omega_{np}^2$, $b_p = k/(J\ell)$, and $b_r = k/J_r$ allows the equations to be written compactly as $\ddot{\theta}_p + a\sin\theta_p = -b_p u$ and $\ddot{\theta}_r = b_r u$. For control design we linearize about the inverted equilibrium $\theta_p = \pi$ using the substitution $\delta\theta_p = \theta_p - \pi$, for which $\sin(\pi + \delta\theta_p) \approx -\delta\theta_p$. Defining the state vector $x = [\delta\theta_p,\ \delta\dot{\theta}_p,\ \delta\theta_r,\ \delta\dot{\theta}_r]^T$, the linearized state-space model is $\dot{x} = Ax + Bu$ with

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ -b_p \\ 0 \\ b_r \end{bmatrix}, \tag{2}$$

which is block-diagonal in the pendulum (rows/columns 1–2) and rotor (rows/columns 3–4) coordinates. Using our identified parameter values $a = \omega_{np}^2 = 72.6\ \mathrm{rad}^2/\mathrm{s}^2$, $b_p = 1.026$, and $b_r = 181.44$ (with ω_{np} taken from the experimental free-swing measurement of Section 3), the open-loop eigenvalues are $\pm\omega_{np} = \pm 8.52$ and a repeated 0 (from the rotor). The pair $\{+\omega_{np},\ 0_{\mathrm{rv}}\}$ contains an unstable pole and a marginally stable pole, so the inverted equilibrium is open-loop unstable, as expected. The controllability matrix $[B\ AB\ A^2B\ A^3B]$ has full rank 4, confirming the system is controllable from the single input u.

Full State Feedback Control with Friction Compensation

1. Controller and Friction-Compensator Development

Two-state feedback uses only the pendulum measurements $(\delta\theta_p,\ \delta\dot{\theta}_p)$, reducing the design to placing the eigenvalues of the 2×2 pendulum block $A_1 = [0\ 1;\ a\ 0]$, $B_1 = [0;\ -b_p]$. We chose closed-loop poles at $-7.5 \pm j13$ ($\omega_n = 15$ rad/s, $\zeta = 0.5$), which satisfies the design constraints $\omega_n > \omega_{np}$, $\zeta < 1/\sqrt{2}$, and keeps each gain magnitude well below 400. Solving for K_2 yields $K_2 \approx [-290.3,\ -14.62]$. In simulation this controller stabilizes the inverted pendulum but, because the rotor receives no velocity feedback, any disturbance on the pendulum arm causes the wheel to accelerate without bound – the rotor pole at zero is unobserved by the controller. Three-state feedback addresses this by additionally feeding back the rotor velocity $\delta\dot{\theta}_r$ while leaving the rotor position $\delta\theta_r$ untouched (its absolute angle is irrelevant to balancing). With closed-loop poles placed at $-7.5 \pm j13$ and a third real pole at -7.5 (at the real part of the complex pair, satisfying the manual's hint that this pole lie between the complex pair and the rotor position pole at 0), the resulting gain is $K_3 \approx [-400.0,\ -44.6,\ 0,\ -0.128]$.

Friction was identified in Section 2 by commanding the motor to constant speeds $\{\pm 25, \pm 50, \pm 75, \pm 100, \pm 150\}$,rad/s under PI velocity control and recording the steady-state control effort, which equals $-F(\dot{\theta}_r)$ at equilibrium. A straight-line fit to the positive- and negative-speed branches gave asymmetric coefficients $b^+ = 0.01178$, $c^+ = 0.5255$ *(forward) and* $b^- = 0.01158$, $c^- = -0.565$ *(reverse)*, which were then implemented in the friction-compensation block $F'(\dot{\theta}_r) = b\dot{\theta}_r + c$ that adds an equal and opposite signal to the motor command. On the hardware, friction compensation provided a clear improvement to the three-state controller. The mechanism is most easily seen by considering what Coulomb friction does to the small commanded corrections the controller produces near the balanced state: the static term c creates a dead-band of width $2c$ around zero motor input, so any command with $|u| < c$ is fully absorbed by friction and no torque reaches the rotor. With friction comp disabled, the controller could not produce the small fine corrections needed to arrest the pendulum's slow drift near $\theta_p = \pi$; instead, it had to wait for the pendulum error to grow large enough that the commanded $|u|$ crossed the dead-band threshold, at which point a sudden large correction was applied. The result was a large, ratcheting limit cycle. Enabling friction compensation cancels $F(\dot{\theta}_r)$ at the motor input, so the controller's commanded u reaches the rotor in full regardless of magnitude – the small corrections actually have small effects – and the limit-cycle amplitude collapses correspondingly.

2. Stability Proof for the Linearized, Frictionless Closed-Loop System

Substituting $u = -K_3 x$ into $\dot{x} = Ax + Bu$ gives $\dot{x} = (A - BK_3)x$. Because $\delta\theta_r$ does not appear in the dynamics of $\delta\theta_p$, $\delta\dot{\theta}_p$, or $\delta\dot{\theta}_r$ – its column in A is zero and K_3 has a zero in the $\delta\theta_r$ slot – we can study stability on the reduced 3-dimensional subspace formed by the states $\delta\theta_p$, $\delta\dot{\theta}_p$, $\delta\dot{\theta}_r$. The reduced model is $\dot{x}_r = A_r x_r + B_r u$ with

$$A_r = \begin{bmatrix} 0 & 1 & 0 \\ a & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad B_r = \begin{bmatrix} 0 \\ -b_p \\ b_r \end{bmatrix}, \qquad K_r = [k_1\ k_2\ k_4] = [-400.0,\ -44.6,\ -0.128]. \tag{3}$$

To locate the equilibria, set $\dot{x}_r = 0$: this gives $\delta\dot{\theta}_p = 0$ (row 1), $a\,\delta\theta_p - b_p u = 0$ (row 2), and $b_r u = 0$ (row 3). From row 3, $u = 0$, and substituting into row 2 yields $a\,\delta\theta_p = 0$, so $\delta\theta_p = 0$. Combined with $\delta\dot{\theta}_p = 0$ and the requirement $u = -K_r x_r = 0 \Rightarrow -k_1\delta\theta_p - k_2\delta\dot{\theta}_p - k_4\delta\dot{\theta}_r = 0$, which forces $\delta\dot{\theta}_r = 0$ since $k_4 \neq 0$. The unique equilibrium of the reduced system is therefore $x_r = [0,0,0]^T$, confirming that the linearized inverted position is the only candidate equilibrium for the closed-loop controller.

Local asymptotic stability follows by examining the characteristic polynomial of $A_r - B_r K_r$. By construction (the `place()` command), its roots are exactly the desired closed-loop poles $\{-7.5 + j13,\ -7.5 - j13,\ -7.5\}$, all of which have strictly negative real parts. Since $A_r - B_r K_r$ is Hurwitz, $x_r(t) \to [0,0,0]^T$ as $t \to \infty$ for every initial condition in a neighborhood of the equilibrium. Equivalently, $\delta\theta_p \to 0$ (i.e. $\theta_p \to \pi$), $\delta\dot{\theta}_p \to 0$, and $\delta\dot{\theta}_r \to 0$, which is precisely the stabilization claim. The rotor position $\delta\theta_r$ is the integral of $\delta\dot{\theta}_r$ and therefore approaches a constant determined by the disturbance history; this is harmless because absolute rotor angle does not enter any other state equation.

3. Simulation Robustness (from Simulink, nonlinear model)

	Two-State (4.2)		**Three-State (4.3)**		**Observer (5.1)**	
Max IC deviation	$\delta\theta_p$ (rad)	$\delta\dot{\theta}_p$ (rad/s)	$\delta\theta_p$ (rad)	$\delta\dot{\theta}_p$ (rad/s)	$\delta\theta_p$ (rad)	$\delta\dot{\theta}_p$ (rad/s)
Max pulse (5% duty, 4 s)	[8]		[8]		[10]	
Max constant disturbance	[3]		[5]		[5]	

Table 1: Maximum disturbance each controller can stabilize, measured in Simulink using the nonlinear RWP block. Pulse and constant disturbances are applied to the pendulum arm (τ_p in the nonlinear model). Fill in the bracketed cells with your simulated values.

4. C2000 Hardware Implementation

When loaded onto the C2000 DAQBoard, the three-state controller with friction compensation exhibited noticeably bimodal behavior. In some trials the controller settled the pendulum to an almost-motionless inverted balance, with a barely-visible limit cycle of a few encoder counts and a rotor that hovered near zero velocity. In other trials, however, the controller would enter a high-amplitude oscillatory regime in which the rotor accelerated rapidly back and forth and shook the entire test fixture, even after appearing to have first achieved balance. We attribute this failure mode to the sensitivity of the discrete encoder-derivative velocity feedback at small rotor speeds: when the pendulum drifts slowly and the quantized $\hat{\dot{\theta}}_r$ estimate carries a large relative error, the controller can interpret a near-zero true velocity as a large false velocity and apply an overcorrecting torque, which then gets reinforced on the next sample. The controller is locally stable in the linearized sense, but the noise of the velocity estimate at near-zero speeds occasionally pushes the system outside the region where that linearization is meaningful.

Full State Feedback Control with Decoupled Observer

1. Why Observers, and Why Decouple

Direct measurements are only available for $\delta\theta_p$ and $\delta\theta_r$ from the two encoders, while the three-state controller requires $\delta\dot{\theta}_p$ and $\delta\dot{\theta}_r$ as well. Although both velocities can be approximated by numerically differentiating the encoder signal, the discrete derivative shown in Section 2 is noisy and the second derivative is essentially unusable. An observer reconstructs the unmeasured states by integrating a copy of the plant model driven by the same input u, with a correction term $L(y - C\hat{x})$ that drives the estimate toward the true state at a rate set by the observer poles. Because the system's A matrix is block-diagonal in the pendulum and rotor coordinates, the four-state observer separates cleanly into two independent 2-state observers – one estimating $(\delta\theta_p,\ \delta\dot{\theta}_p)$ from $y_1 = \delta\theta_p$, the other estimating $(\delta\theta_r,\ \delta\dot{\theta}_r)$ from $y_2 = \delta\theta_r$. Compared with a single 4-state observer designed via `place()` on the full A, the decoupled design (i) zeroes out the off-block-diagonal entries of L that are mathematically unnecessary, and (ii) avoids the spurious sensitivity those off-diagonal entries create with respect to model error and encoder noise. We observed exactly this in Section 5.1: the 4-state observer was extremely sensitive to small variations in the nonlinear model parameters, while the decoupled 2-plus-2 design was substantially more robust.

2. Proof of Observer Error Convergence

Define the estimation error $e = x - \hat{x}$. The plant evolves as $\dot{x} = Ax + Bu$ and the observer as $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}) = A\hat{x} + Bu + LC(x - \hat{x})$. Subtracting,

$$\dot{e} = \dot{x} - \dot{\hat{x}} = Ax - A\hat{x} - LC(x - \hat{x}) = (A - LC)\,e. \tag{4}$$

The error dynamics depend only on $(A-LC)$ – neither u nor x appears explicitly. Setting $\dot{e}=0$ gives $(A-LC)\,e=0$. For the decoupled design, $A-LC$ takes the block-diagonal form

$$(A-LC)=\begin{bmatrix} A_1-L_1C_1 & 0 \\ 0 & A_2-L_2C_2 \end{bmatrix}, \tag{5}$$

with $A_1=[0\ 1;\ a\ 0]$, $C_1=[1\ 0]$, $L_1=[l_{11};\ l_{21}]$ for the pendulum, and $A_2=[0\ 1;\ 0\ 0]$, $C_2=[1\ 0]$, $L_2=[l_{32};\ l_{42}]$ for the rotor. The observer gains were chosen with the pendulum subsystem poles placed at $\{-90,\,-95\}$ and the rotor subsystem poles at $\{-100,\,-105\}$, all roughly $6\times$ faster than the closed-loop pendulum poles at real part -7.5. This gives $L_1=[185;\ 8622.6]$ and $L_2=[205;\ 10500]$, with characteristic polynomials $(s+90)(s+95)=s^2+185s+8550$ for the pendulum block and $(s+100)(s+105)=s^2+205s+10500$ for the rotor block. Because every eigenvalue of $A-LC$ lies in $\{-90,\,-95,\,-100,\,-105\}$, all strictly negative, $\det(A-LC)=8550\cdot 10500\neq 0$, so the only solution of $(A-LC)\,e=0$ is $e=0$, and since $A-LC$ is Hurwitz, $e(t)\to 0$ exponentially fast for any initial estimation error. Equivalently, $\hat{x}(t)\to x(t)$, so the observer-based controller asymptotically recovers the performance of the (impractical) full-state feedback law that has direct access to all four states.

3. Simulation Robustness

Robustness values measured in Simulink for the observer-based controller are included in the rightmost columns of Table 1 above. Because the controller now consumes the observer's estimates rather than the true model states, the achievable disturbance amplitudes are somewhat smaller than for the idealized three-state design – the gap is the price paid for using estimates that lag the true states by $O(1/90)\approx 11\,\text{ms}$ during transients.

4. C2000 Hardware Implementation

On the actual hardware, the observer-based controller behaved qualitatively differently from the direct three-state design: it was substantially more consistent. The observer never reached the near-zero-error condition that the direct three-state controller occasionally achieved, but it also never entered the violent oscillatory regime that the direct controller fell into on roughly half of our trials. Instead, the observer-based controller produced a steady, modest limit cycle and recovered cleanly from small taps on the pendulum arm. We interpret this as a direct consequence of the observer's filtering action: by integrating the plant model forward and correcting it slowly with the encoder measurements, the observer suppresses the high-frequency quantization noise that destabilized the direct controller, at the cost of slightly slower transient response. The dynamics are smoother because the velocity estimate is smoother.

Conclusions

We define performance along two axes: the size of the disturbance region each controller can tolerate (Table 1) and the *consistency* of the inverted balance on hardware (repeatability of recovery, prevalence of failure modes). By the first measure, the direct three-state controller produced slightly larger tolerable disturbances in simulation. By the second measure, however, the observer-based controller was clearly better: it balanced the pendulum reliably on essentially every trial, while the direct three-state controller exhibited bimodal behavior – occasionally settling to a near-zero-error balance, but in many trials entering a violent self-reinforcing oscillation that shook the entire fixture. We judge consistency the more important criterion in a real implementation, since a controller that is "best when it works" but unpredictable in when that is would not be acceptable for any application where the cost of failure matters. On that basis, the observer-based controller performed better overall.

The underlying reason is that the direct three-state controller relies on a discrete derivative of the encoder signal, which is extremely noisy at the small rotor speeds typical of the balanced condition; the observer instead reconstructs $\dot{\theta}_r$ from a forward-integrated plant model corrected by encoder position, which filters that noise. The price paid is slightly slower transient response from the observer, since its estimates lag the true state by roughly the inverse of the observer pole frequency. We were willing to pay that price for the dramatic improvement in reliability.

Appendix A: Simulink Diagrams

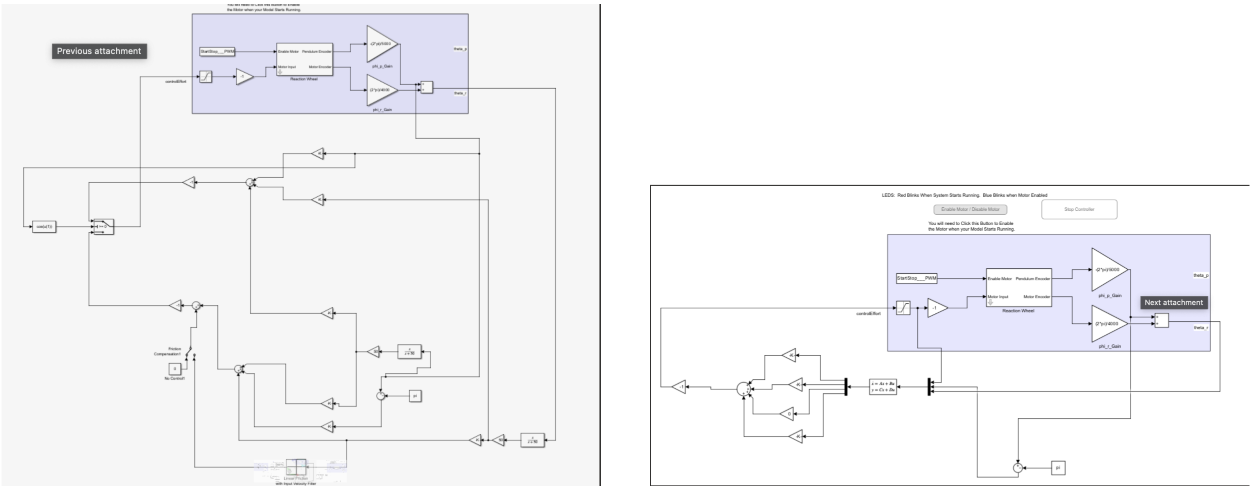

Figure 1: Left:2 three-state controller that makes our up down stabilizing control. Right: observer-based controller.

Appendix B: Optional Sections (Extra Credit)

Section 6: Up and Down Stabilizing Control

We extended the three-state feedback design of Section 4.3 into a switching controller that stabilizes the pendulum in *both* the inverted and hanging configurations. The structure consists of two parallel three-state feedback laws and a Simulink Switch block that selects between them in real time.

The two controllers differ only in which equilibrium they linearize about. For the up controller, $\sin(\pi+\delta\theta_p) \approx -\delta\theta_p$ gives $A_{\text{up}} = [0\ 1\ 0\ 0;\ a\ 0\ 0\ 0;\ 0\ 0\ 0\ 1;\ 0\ 0\ 0\ 0]$, which is the design from Section 4.3 and produces $K_{\text{up}} \approx [-400.0,\ -44.6,\ 0,\ -0.128]$. For the down controller, $\sin(\delta\theta_p) \approx +\delta\theta_p$ flips the sign of the entry on the second row, giving $A_{\text{down}} = [0\,1\,0\,0;\ -a\,0\,0\,0;\ 0\,0\,0\,1;\ 0\,0\,0\,0]$. Using the same closed-loop pole locations $\{-7.5 \pm j13,\ 0,\ -7.5\}$, MATLAB's `place` command yields $K_{\text{down}} \approx [-258.4,\ 0.75,\ 0,\ 0.128]$. The opposite sign of k_4 between the two controllers reflects the fact that the rotor must accelerate in opposite directions to damp deviations from $\theta_p = \pi$ versus $\theta_p = 0$.

The switching decision uses $\cos(\theta_p)$ as a region indicator: $\cos(\theta_p) \geq 0$ exactly when the pendulum lies in the half-plane $|\theta_p| < \pi/2$ (closer to the downward equilibrium), while $\cos(\theta_p) < 0$ corresponds to the half-plane closer to the inverted equilibrium. Routing the encoder signal through a `cos(u(1))` Fcn block followed by a ≥ 0 Switch automatically hands control to whichever gain is appropriate for the current pendulum half. Each branch also applies the correct state offset (subtracting π for the up controller; using θ_p directly for the down controller) so that both feedback laws operate on properly-centered delta states. The shared friction-compensation block from Section 4.4 remains downstream of the switch, so it is active for both controllers.

On hardware, the pendulum can now be released from any starting position: when released below the horizontal, the down controller damps out residual swinging substantially faster than the open-loop case; when released above the horizontal, the up controller catches it and balances inverted. The switching transient at $|\theta_p| = \pi/2$ is small in practice because both controllers produce only modest control effort near that crossover point, and neither controller has any wind-up state that would survive the switch.

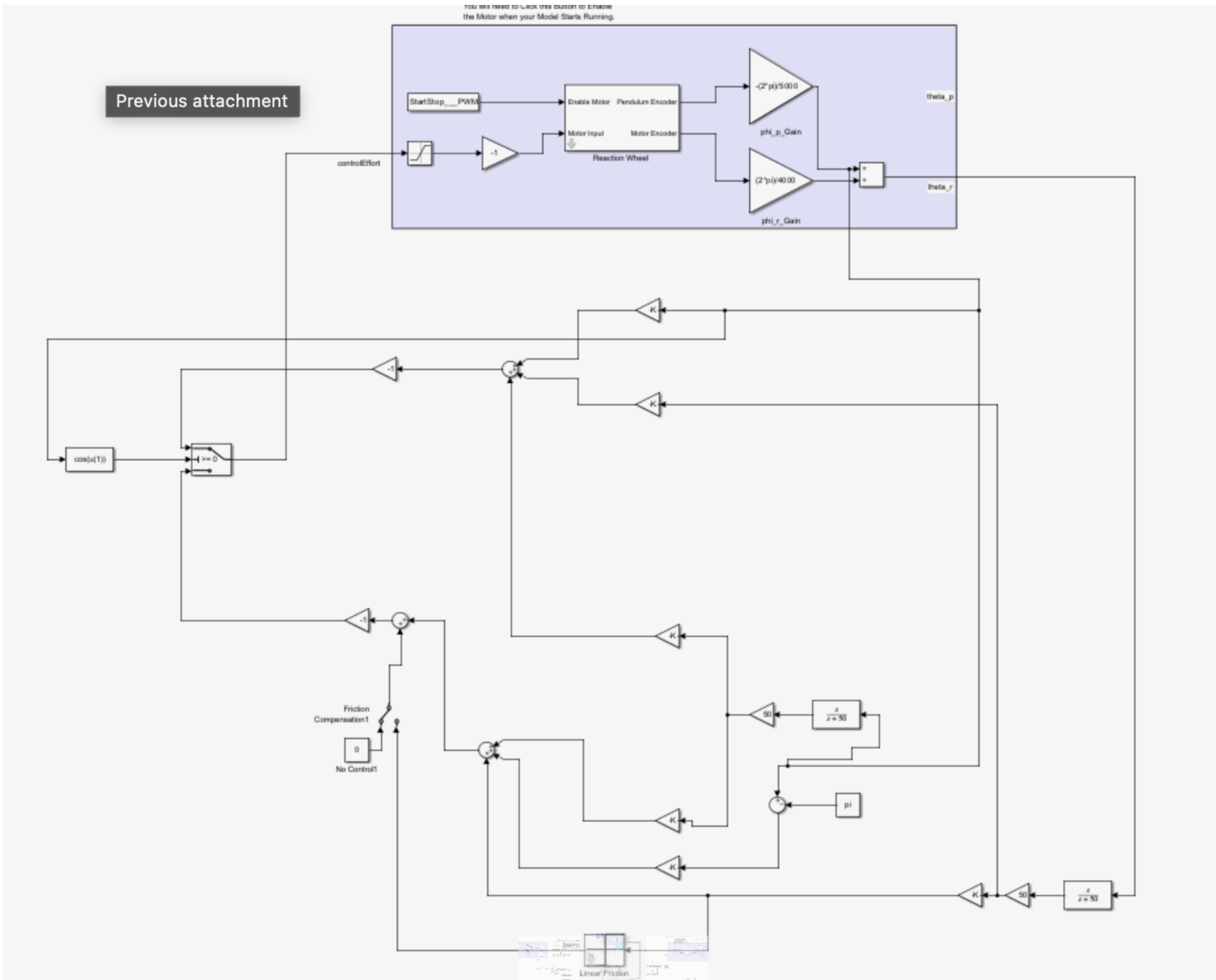

Figure 2: Section 6 switching controller Simulink model. The `cos(u(1))` block at left, combined with the ≥ 0 Switch, selects between K_{up} and K_{down} based on which equilibrium the pendulum is closer to. Velocity estimates use the $s/(s+50)$ continuous derivative approximation from Section 2.1, and the friction compensation branch (bottom) is shared between both controllers.